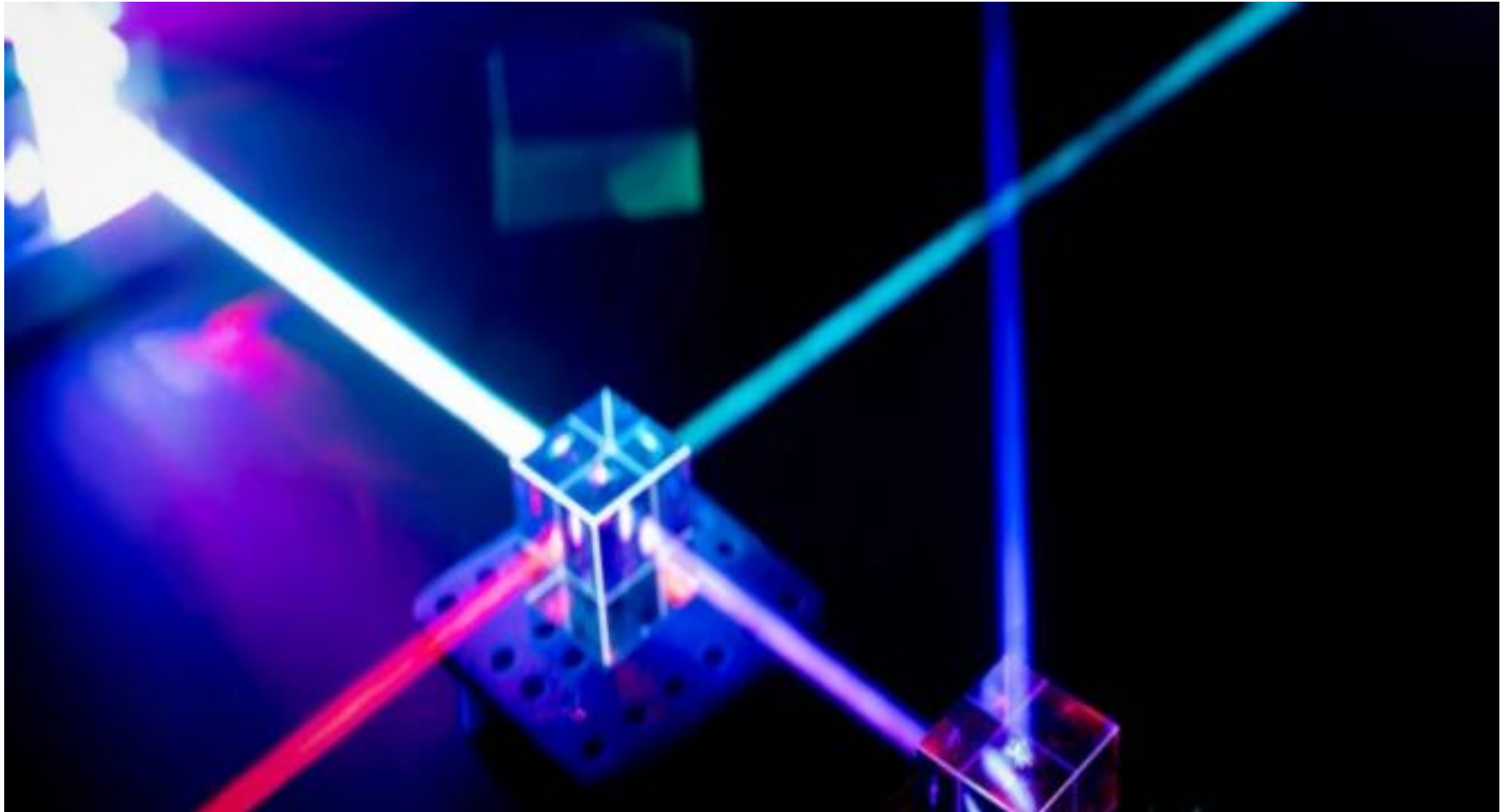
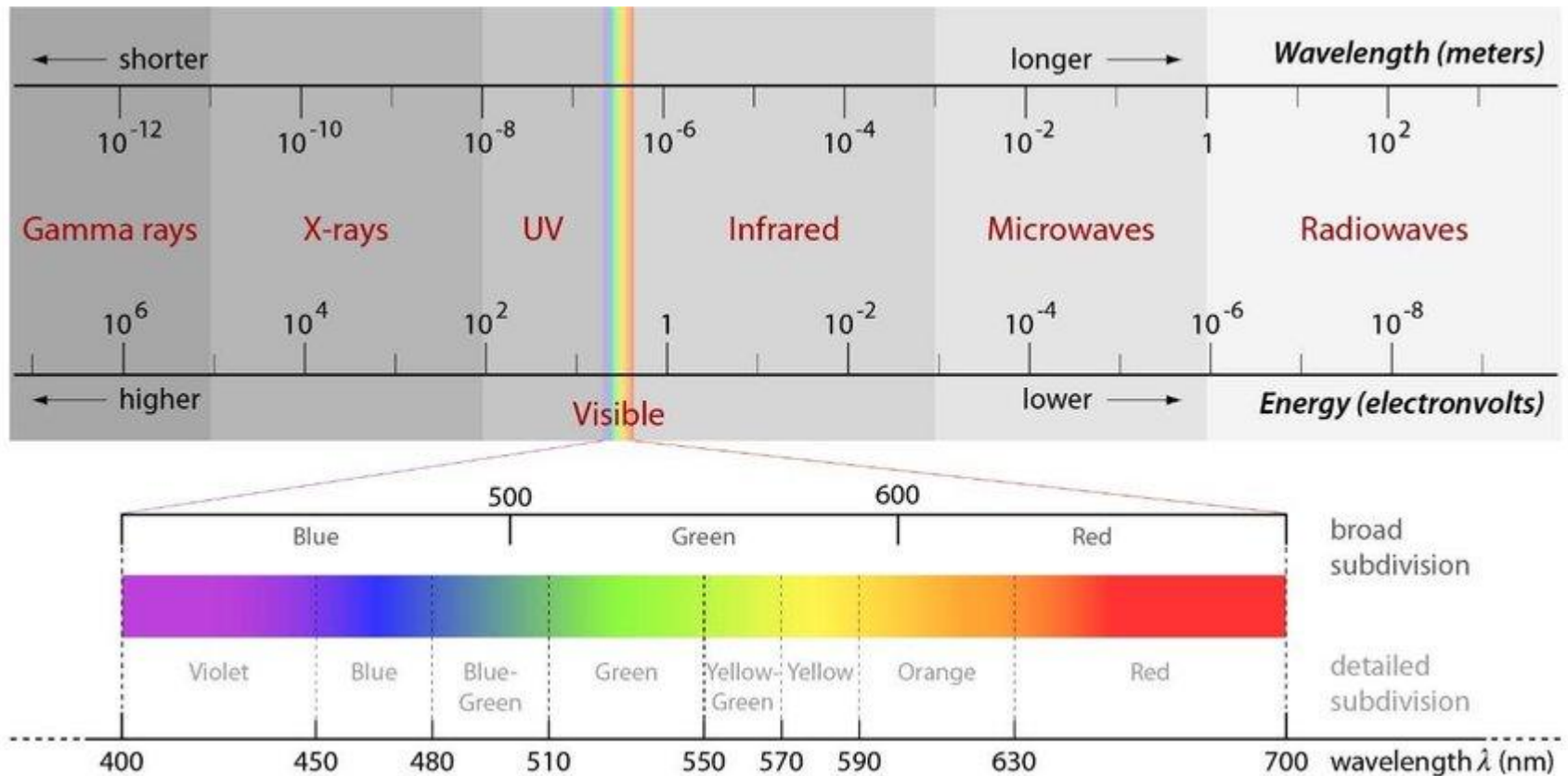


OPTICS AND LASERS

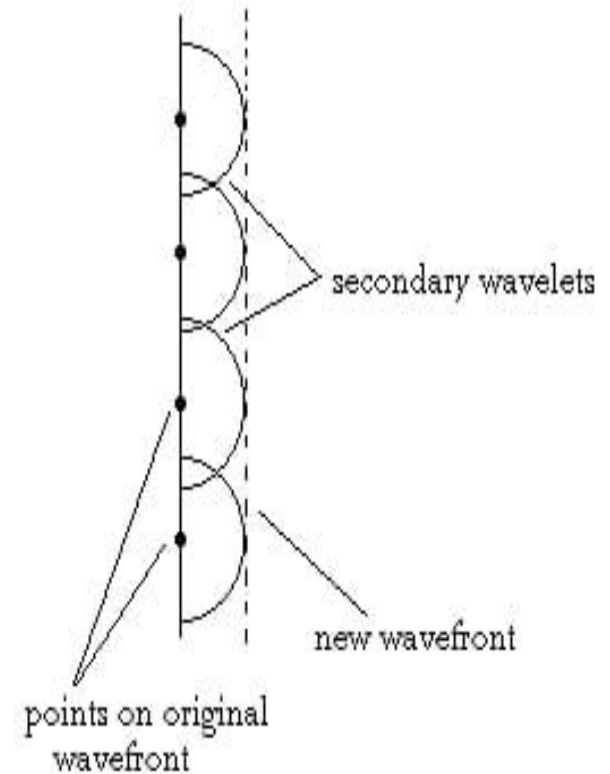
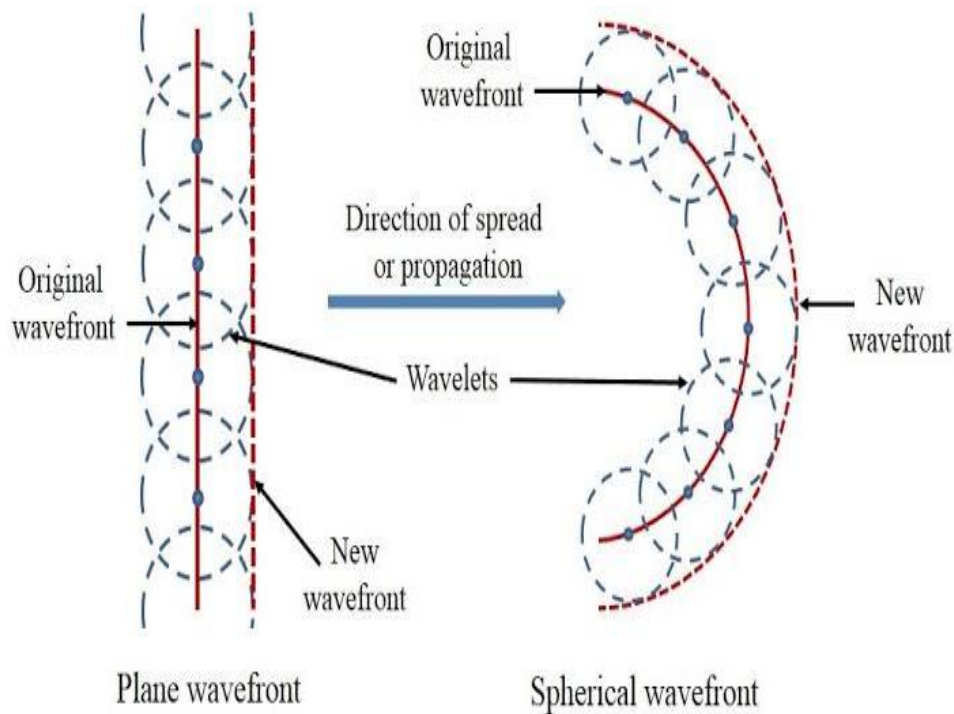


EMR Spectrum:



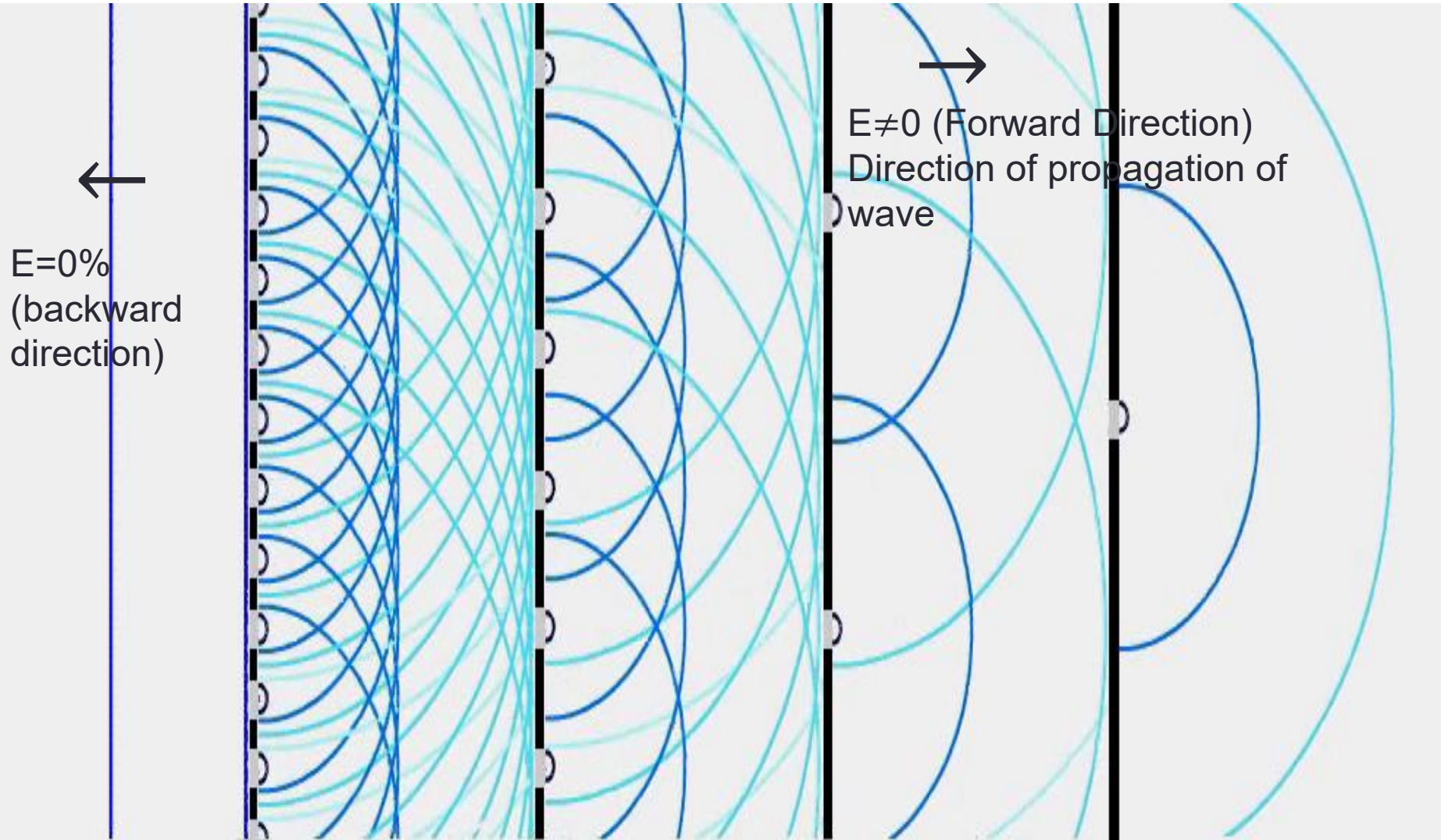
HUYGEN'S PRINCIPLE :

‘Every point on a wave-front acts as a source for secondary waves.’



Huygens principle explained the phenomenon of reflection, refraction and interference.

HUYGEN'S-FRESNEL CORRECTION:



DIFFRACTION:

The bending or speeding of waves around the corners of an obstacle or through an aperture into the region of geometrical shadow of the obstacle/aperture.

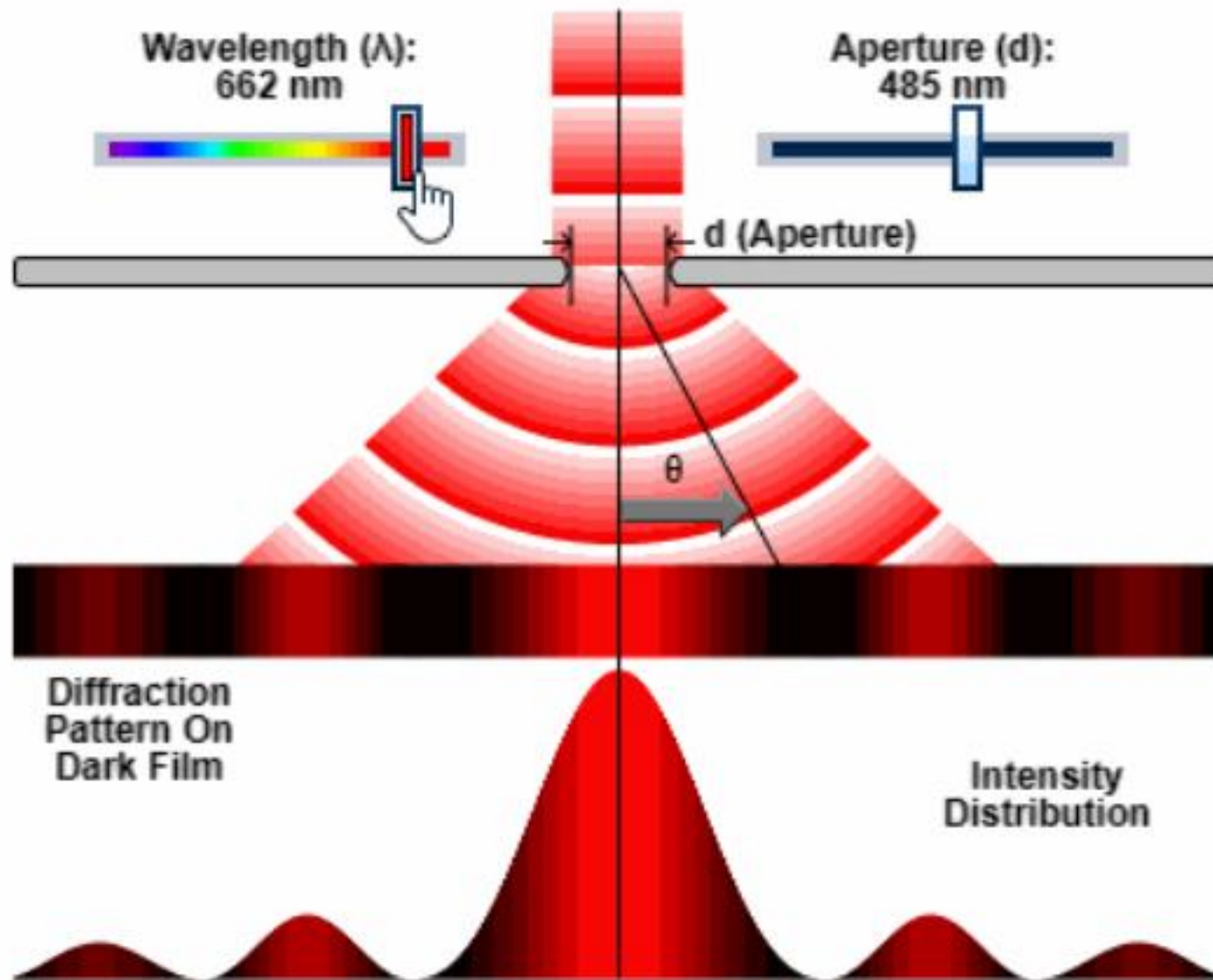
Diffraction refers to phenomena that occur when a wave encounters an obstacle or opening.

$$d \sin \theta = n \lambda$$

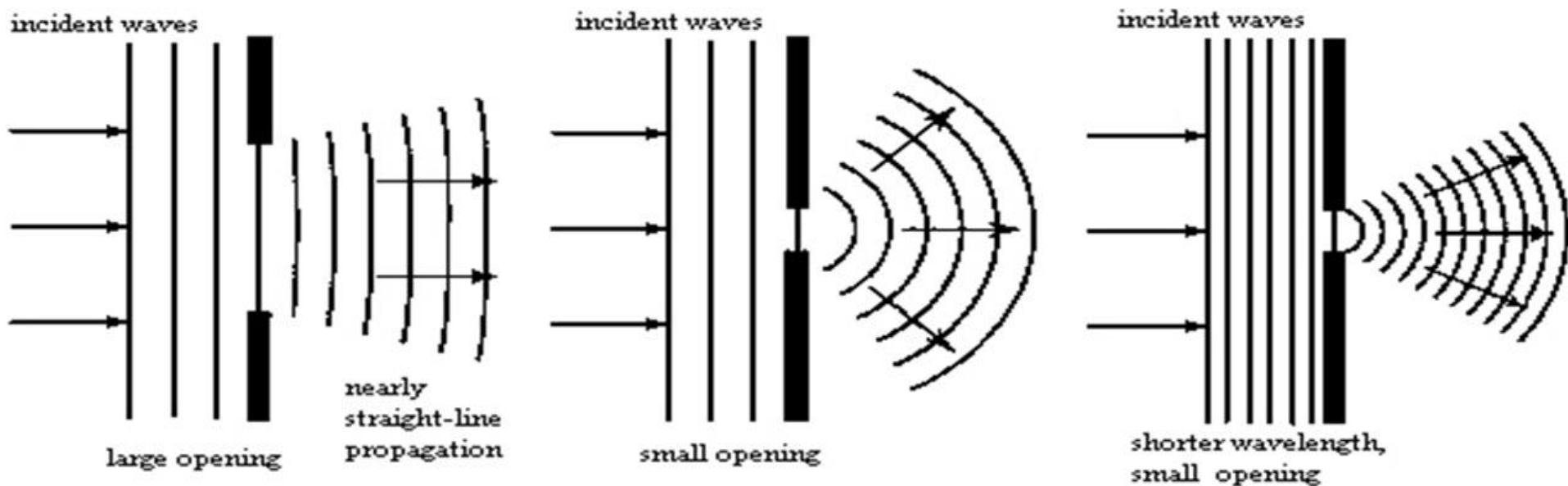
d = distance b/w slits (slit spacing), θ = diffraction angle,

n = order number of maxima, λ = wavelength

Diffraction of Light



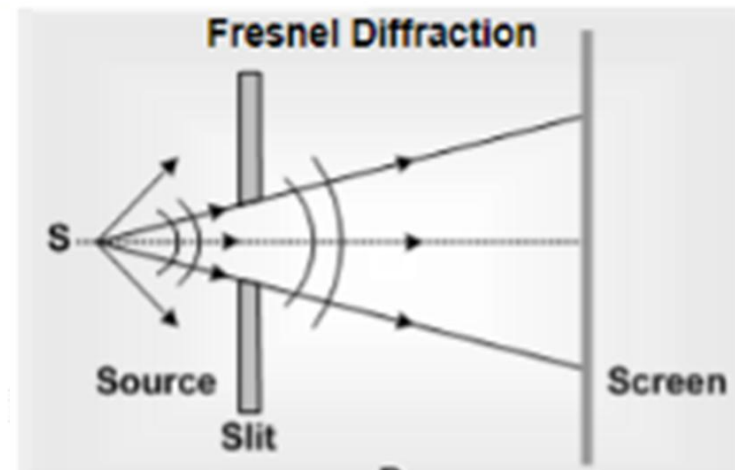
- Quality of diffraction changes with the change in the size of the slit.
- For ideal diffraction, the dimension of the slit are comparable to the wavelength of the light.



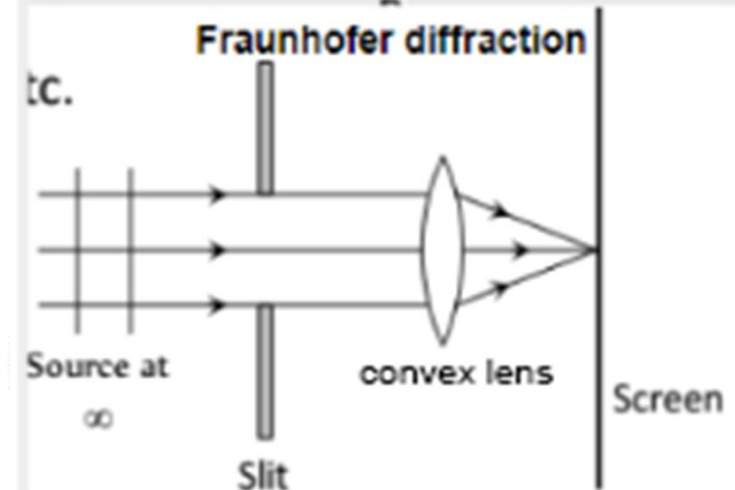
Diffraction

The diffraction phenomenon is of two types:

1) *Fresnel diffraction:* The source of light and the screen are at finite distance from the diffracting aperture.



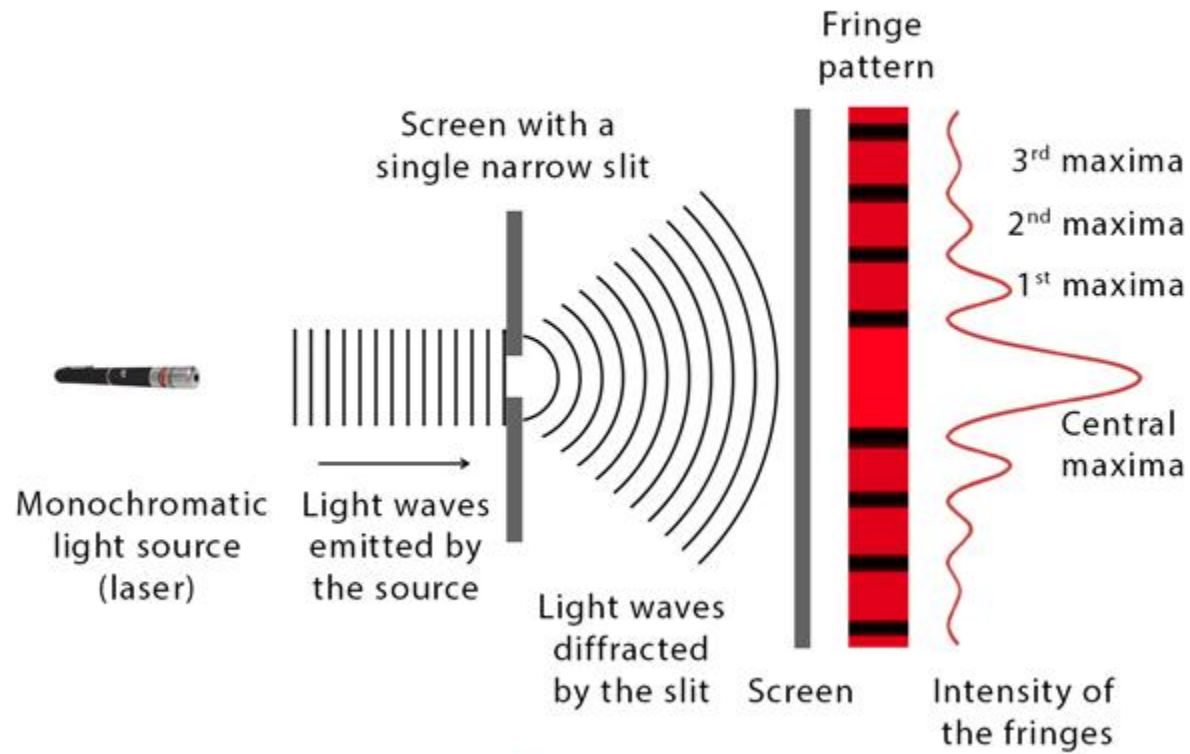
2) *Fraunhofer diffraction:* The source and the screen are at infinite distance from the aperture (Using convex lens)



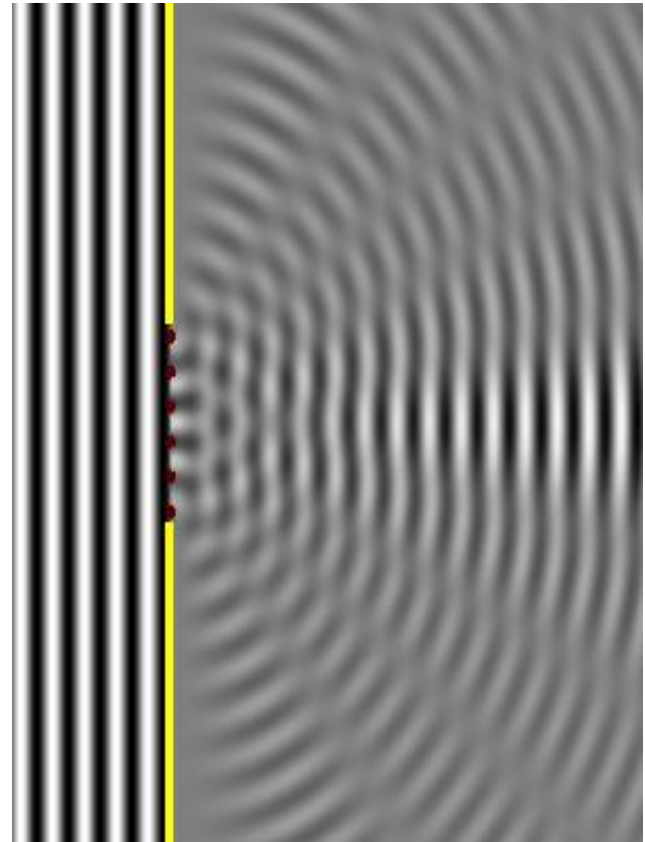
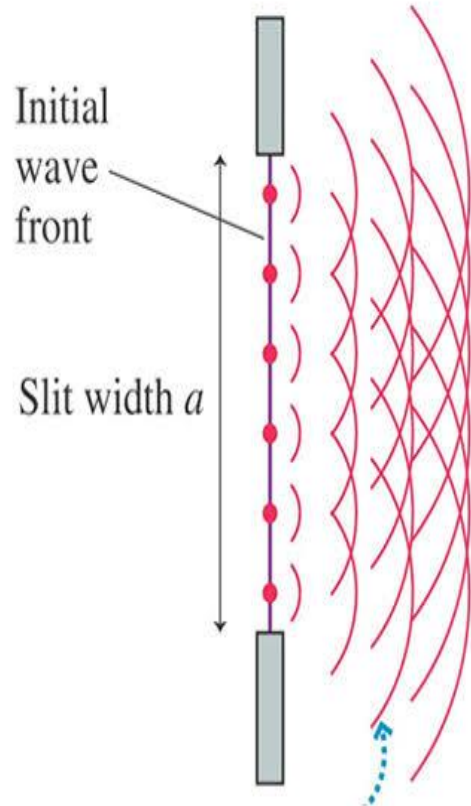
Note: Wavelength of light must be comparable to the size of an obstacle

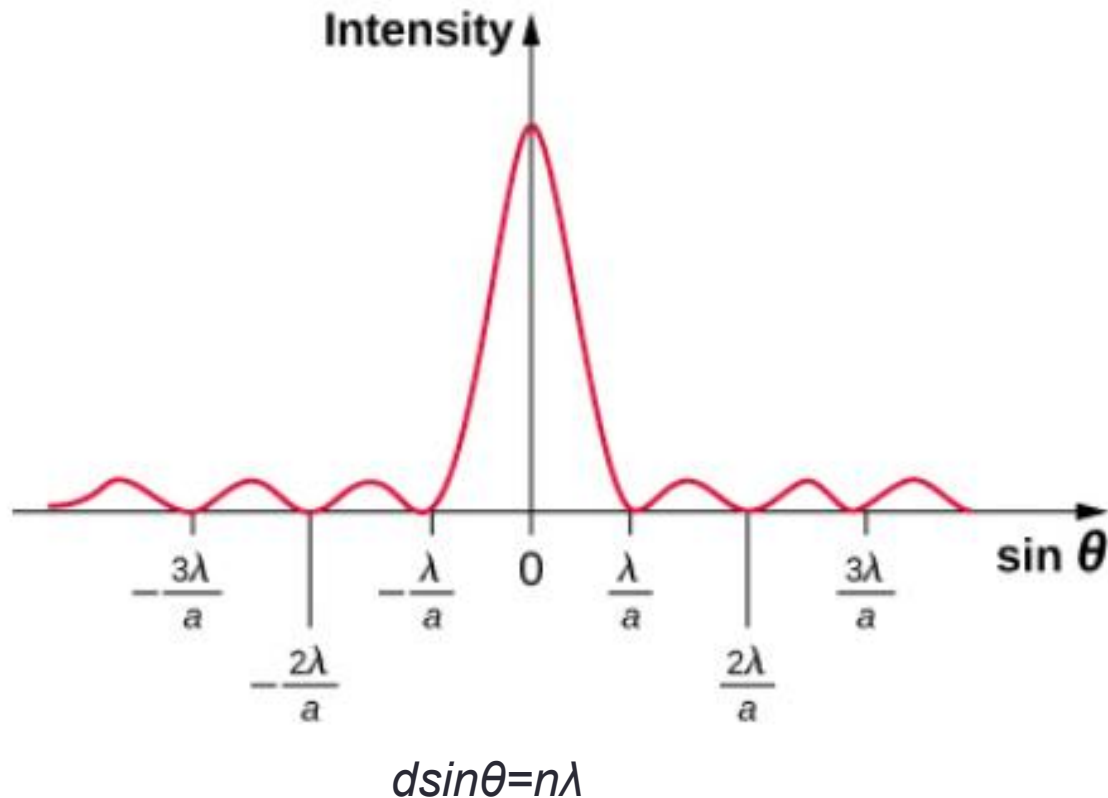
SINGLE SLIT DIFFRACTION:

Single-Slit Diffraction



Single Slit Diffraction





where

$m = \pm 1, \pm 2, \pm 3, \dots$,

a = slit width,

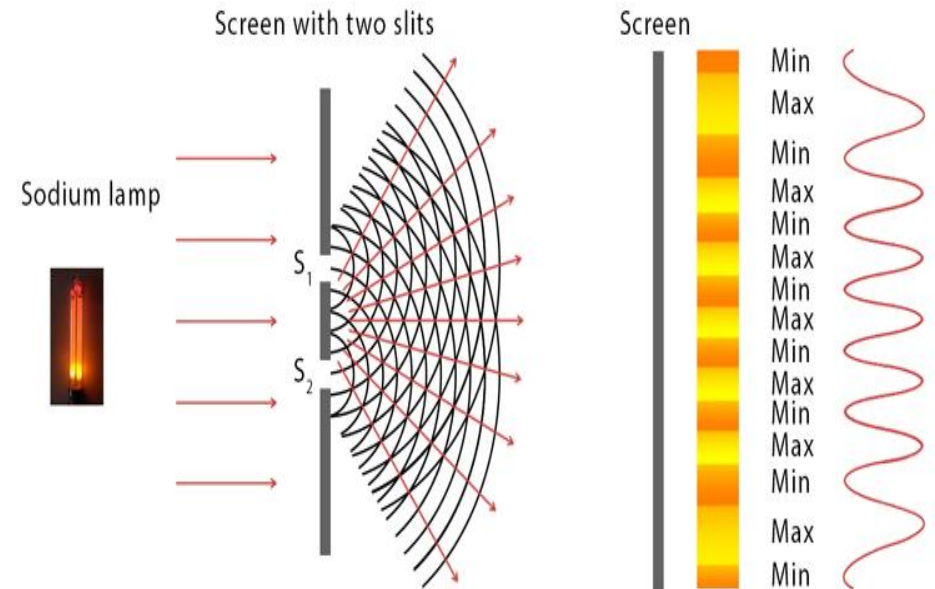
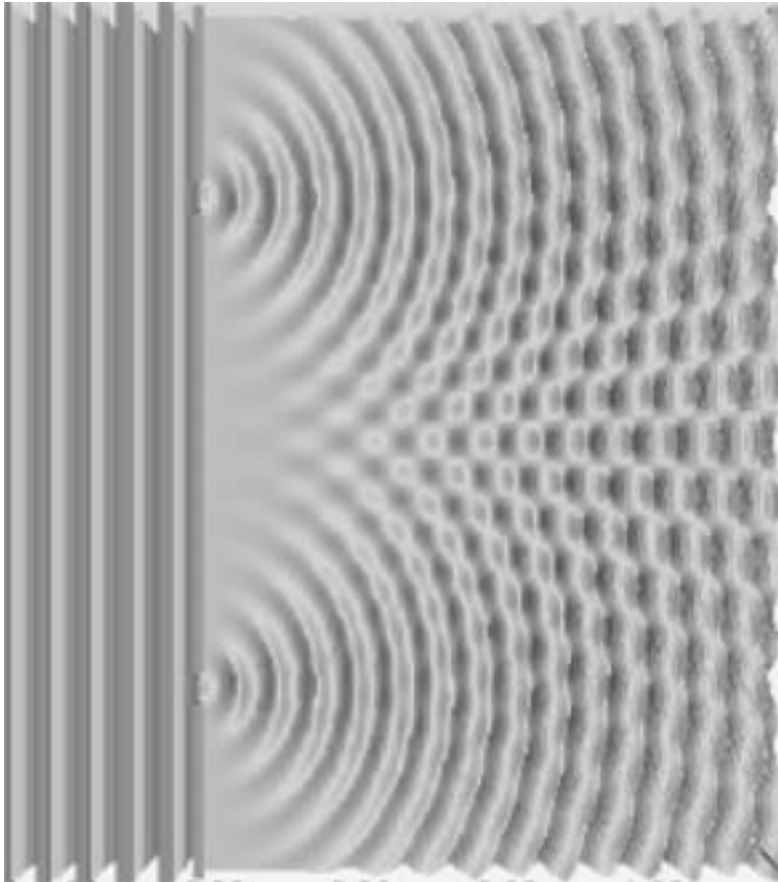
λ = light's wavelength,

θ = angle relative to the original direction of the light, and

m is the order of the minimum.

Graph of intensity for single-slit interference, and it is apparent that the maxima on either side of the central maximum are much less intense and not as wide.

Young's Double Slit Experiment



Light source	Rays of light coming from the source reach the slits	Interference of light waves due to two tiny slits and arrows indicate direction of wave propagation	Alternating bright and dark fringes due to interference of light waves	Intensity of the fringes shows the maxima and minima
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- **constructive interference for a double slit**, the path length difference must be an integral multiple of the wavelength, or

$$d\sin\theta = m\lambda,$$

for $m=0, 1, -1, 2, -2, \dots$ (constructive)..

- **destructive interference for a double slit**, the path length difference must be a half-integral multiple of the wavelength, or

$$d\sin\theta = m + \frac{1}{2} \lambda,$$

for $m=0, 1, -1, 2, -2, \dots$ (destructive)

- Where,

λ = is the wavelength of the light,

d = is the distance between slits, and

θ = is the angle from the original direction of the beam

PRACTICE PROBLEM:

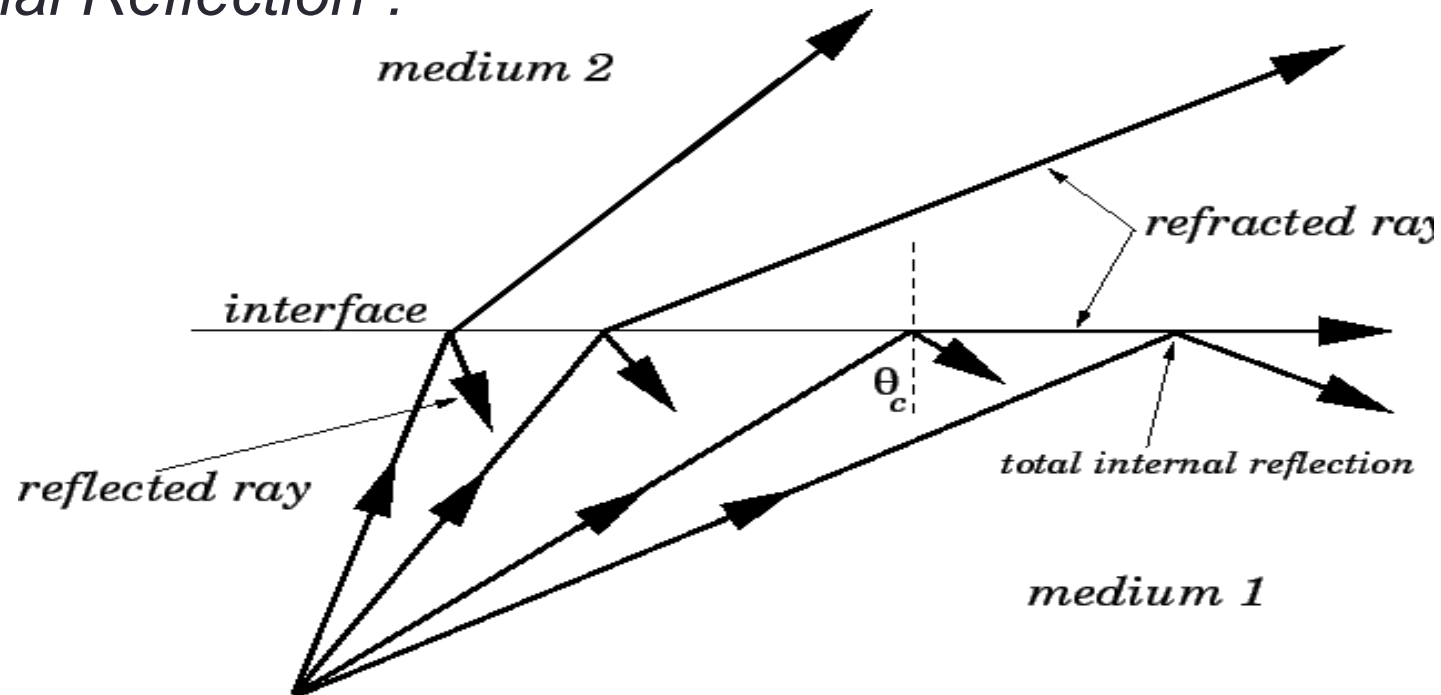
A viewing screen is separated from a double-slit source by 1.2 m. The distance between the two slits is 0.030 mm and the second-order bright fringe is 4.5 cm from the central fringe. Calculate the distance between adjacent bright fringes.

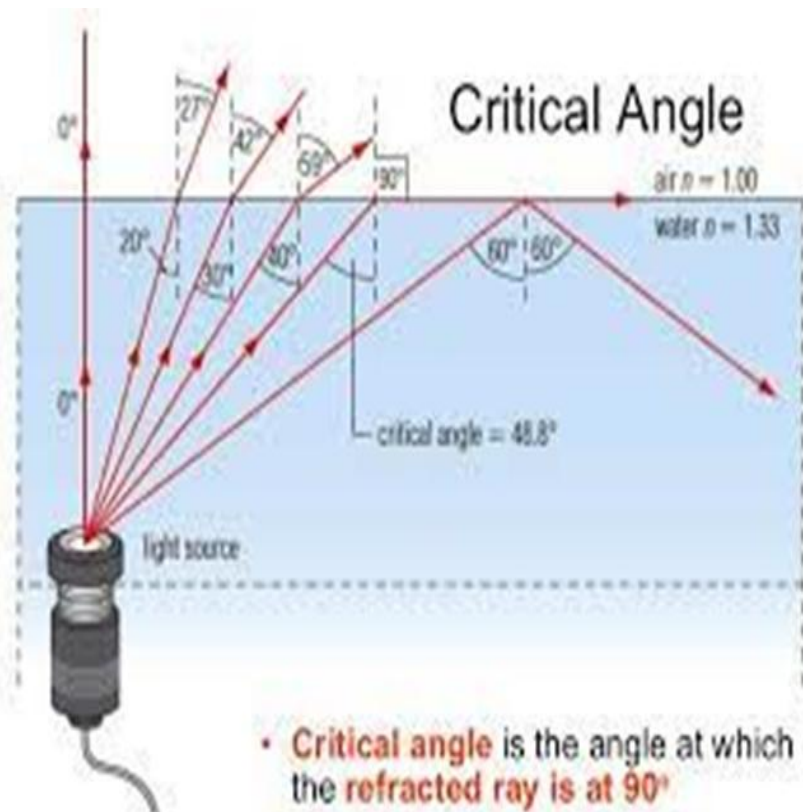
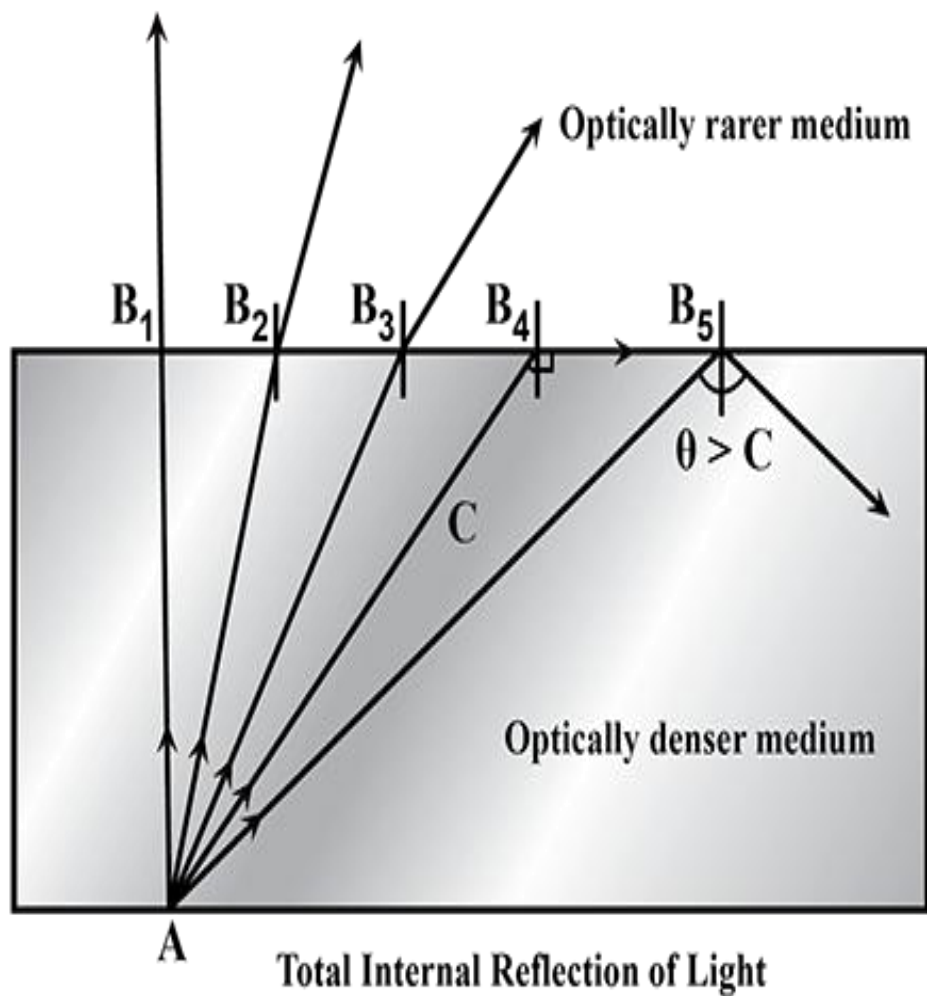
Total Internal Reflection:

If light passes from optically denser to rarer medium, then at interface, the light is partly reflected back into denser medium and partly refracted into rarer medium.

This reflection is called *Total Internal Reflection*.

Under certain conditions, the whole of the incident light can be made to be reflected back into denser medium causing “*Total Internal Reflection*”.





Refraction

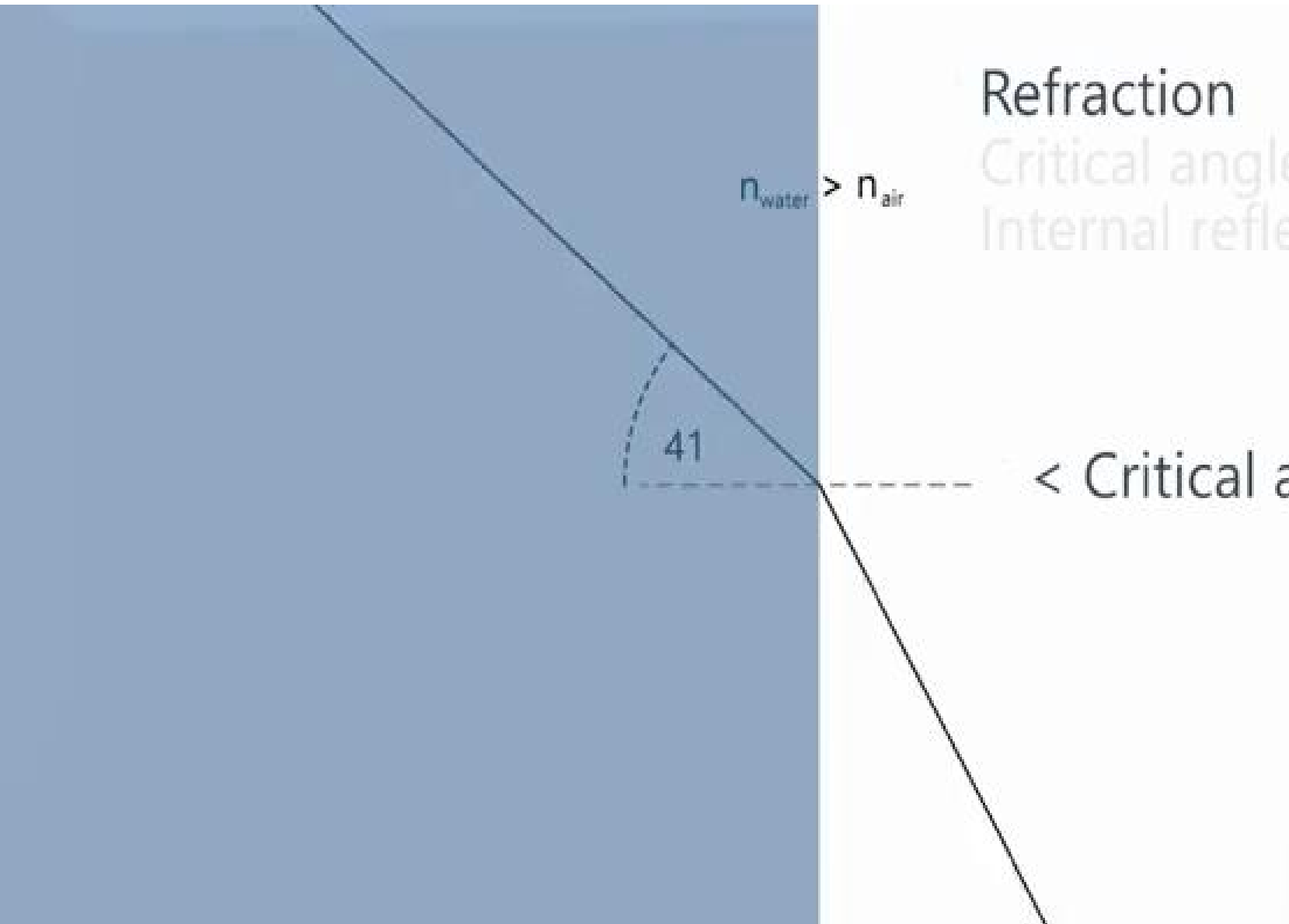
Critical angle

Internal reflection

$$n_{\text{water}} > n_{\text{air}}$$

41

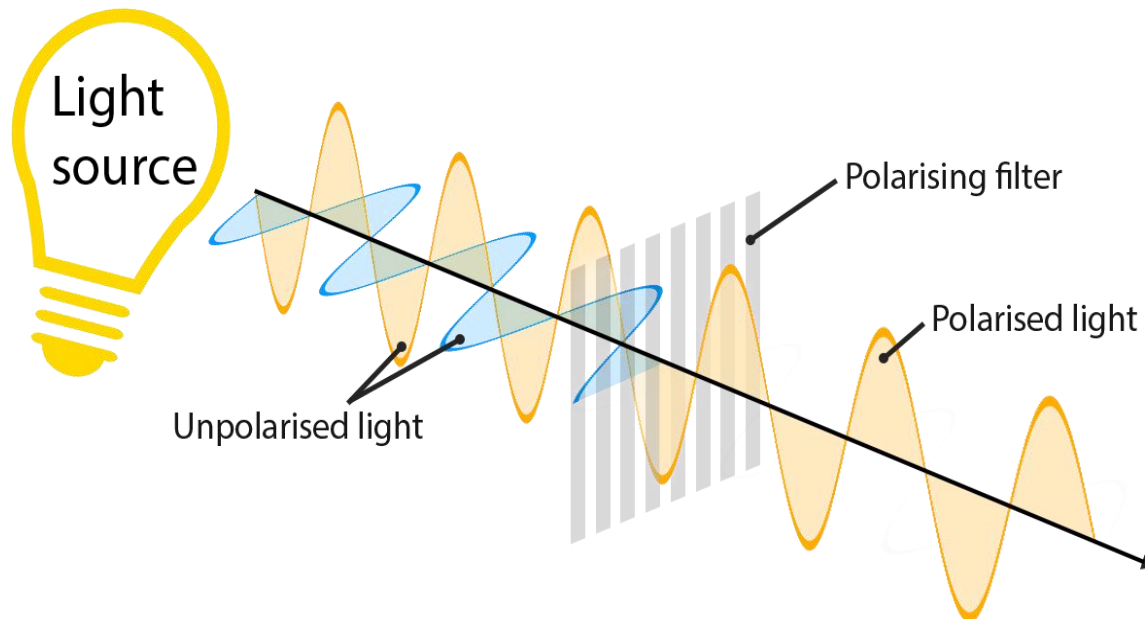
< Critical angle



POLARIZATION:

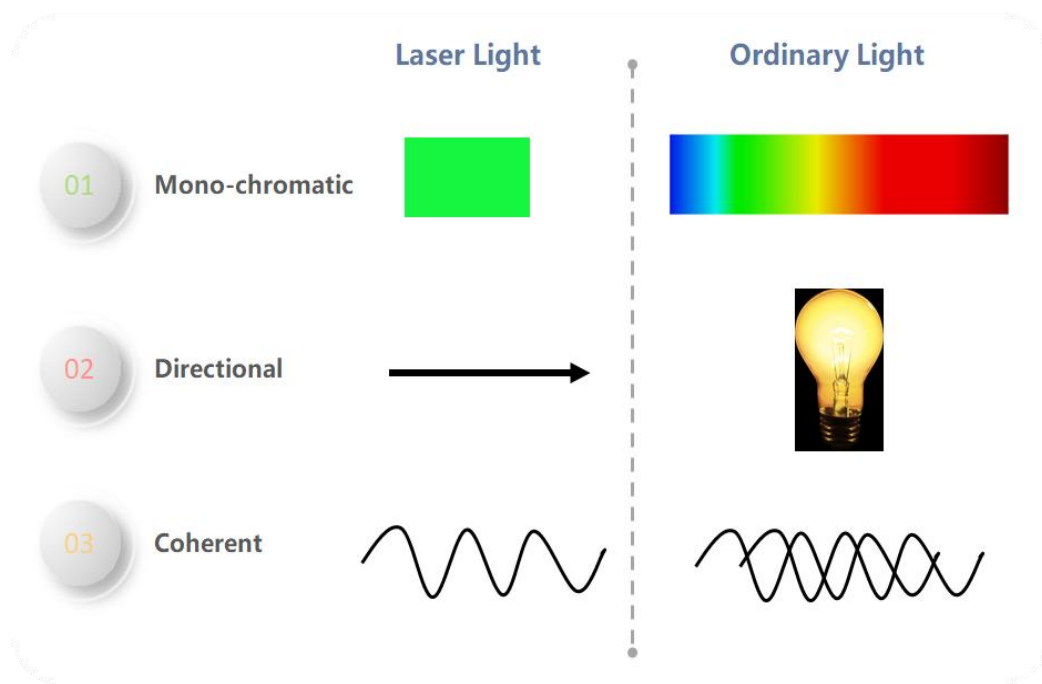
Polarization is a property applying to transverse waves that specifies the geometrical orientation of the oscillations. In a transverse wave, the direction of the oscillation is perpendicular to the direction of motion of the wave.

Light may be polarized by reflection or by passing it through filters, such as certain crystals, that transmit vibration in one plane but not in others.



LASER: Light Amplification by Stimulated Emission of Radiation

Laser is a coherent and monochromatic (consisting single color/ wavelength) light in which all particles (photons) move in one direction and reinforce each other.

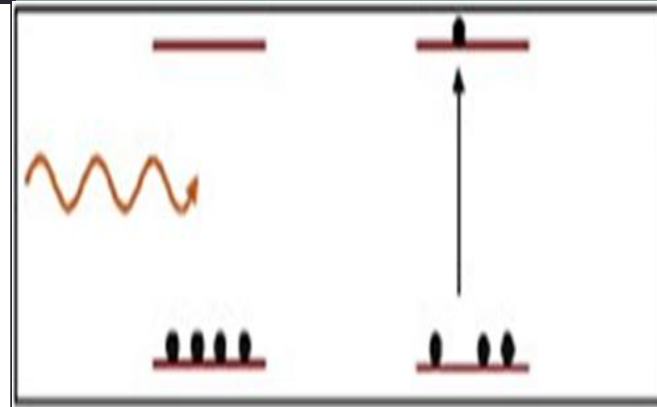


Whereas, ordinary light is incoherent and polychromatic (consisting of several colors/ wavelengths) light.

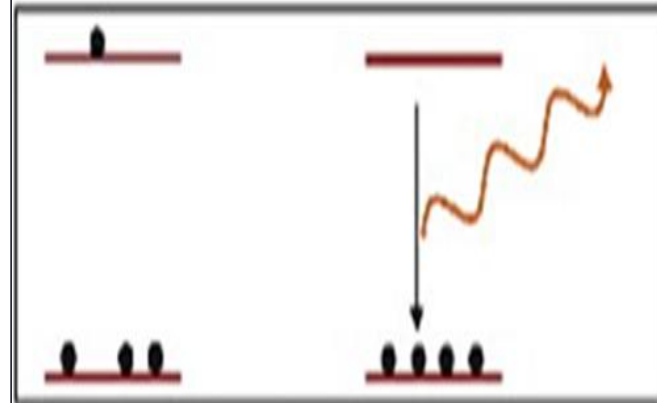
Absorption: The process in which an atom in a lower energy level (E_1) absorbs a photon of frequency " $h\nu$ " and moves to an upper energy level (E_2).

Spontaneous Emission: When an atom in an upper energy level (E_2) decays spontaneously to lower energy level (E_1) and emits a photon (of light) of frequency " $h\nu$ ". The emission is called **spontaneous emission**, the photon emitted has random direction and phase.

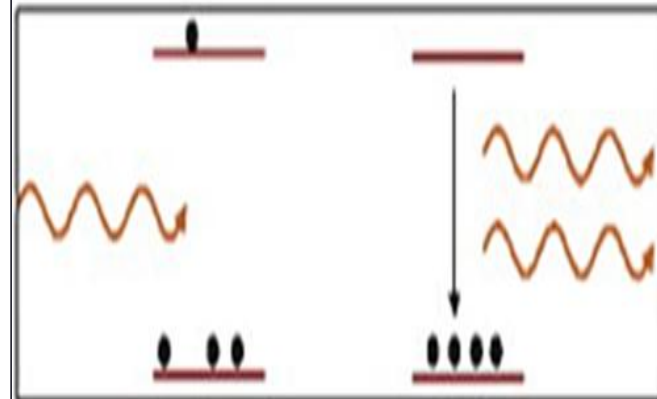
Stimulated Emission: When an incident photon causes an atom in an upper energy level (E_2) to decay, emitting a "stimulated photon" whose properties are identical to those of the incident photon. The emission is called **stimulated emission**, the stimulated emission of photons are in-phase and amplified.



Absorption



Spontaneous Emission



Stimulated Emission

Absorption



Spontaneous Emission



Stimulated Emission



ALYSS



Population Inversion:

Consider a group of electrons with two energy levels E_1 and E_2 .

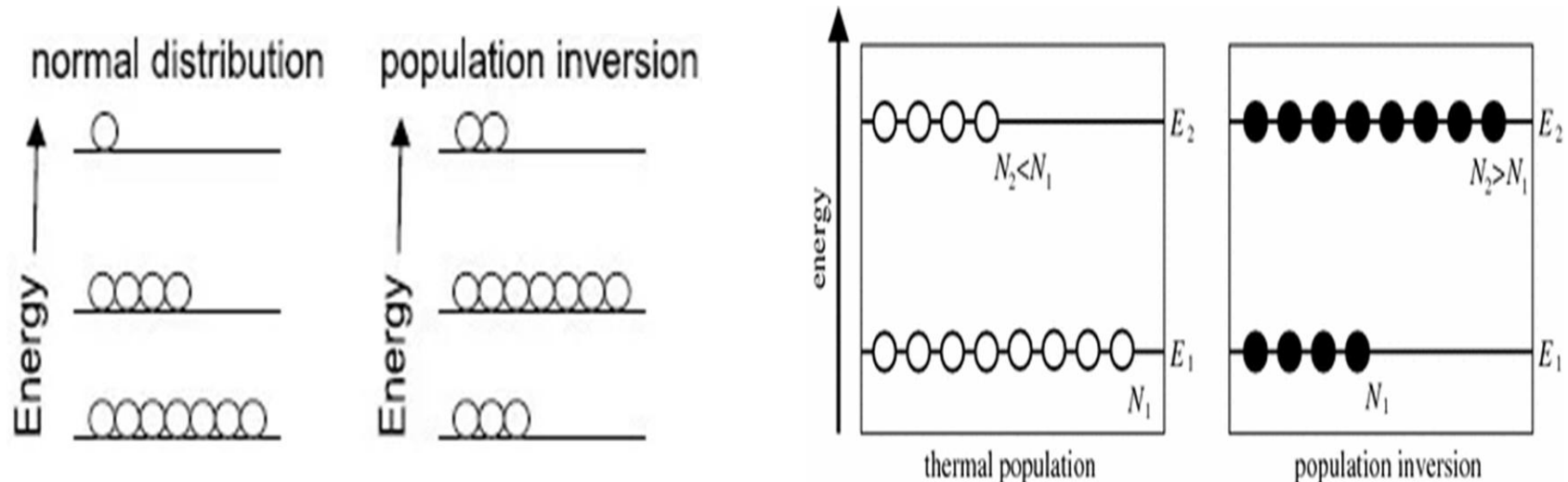
E_1 = lower energy state , E_2 = higher energy state.

N_1 = number of electrons in E_1 .

N_2 = number of electrons in E_2 .

The number of electrons per unit volume in an energy state is the population of that energy state.

- Under normal conditions, $N_1 > N_2$
- When temperature increases, At best, $N_1 = N_2$, which results in no optical gain.
- Therefore, 3 or more energy states to achieve population inversion.
- The greater is the number of energy states the greater is the optical gain.
- For Lasing Action; $N_3 > N_1$ (N_3 =no. of electrons in metastable state).



Three-Level and Four-Level Laser Energy Diagrams

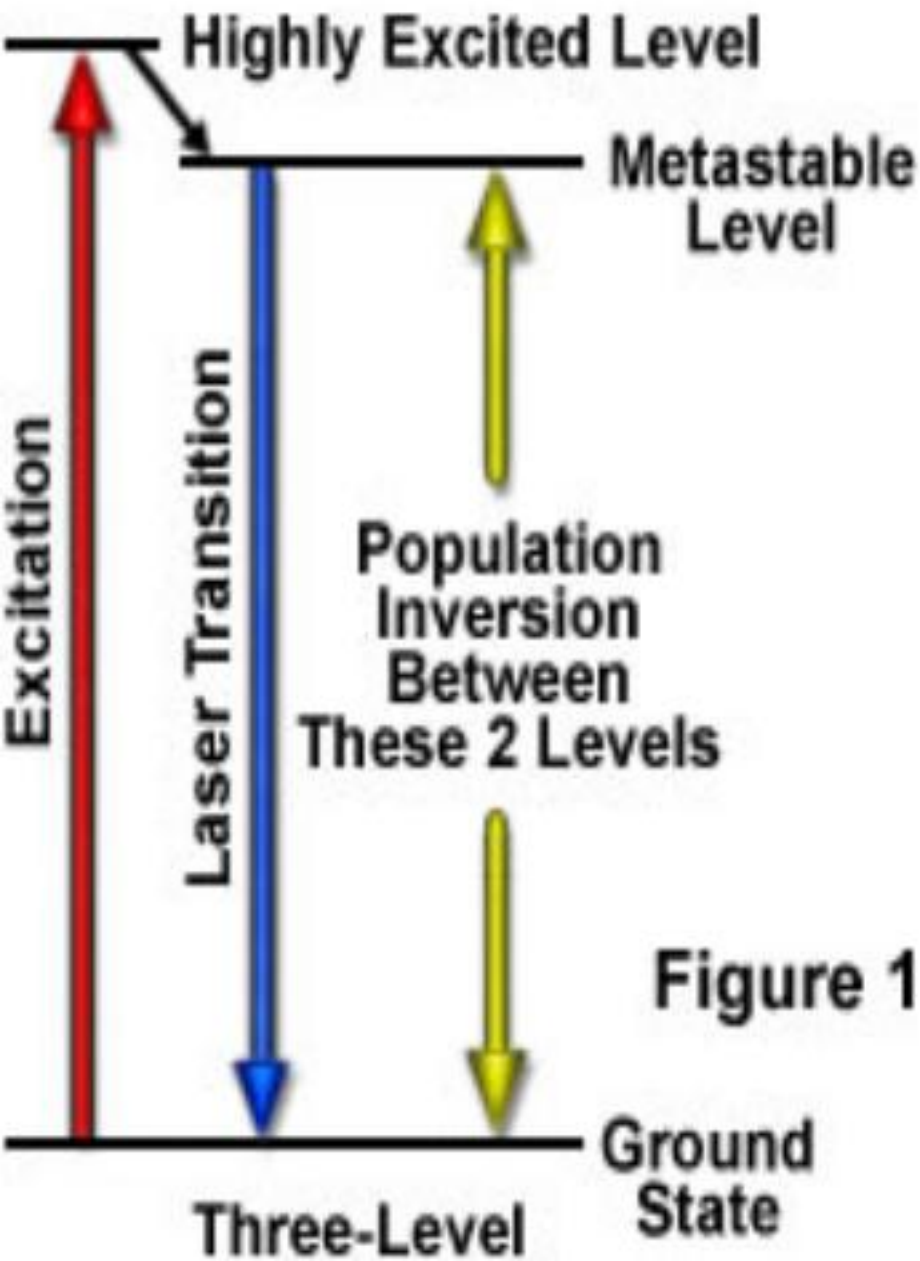
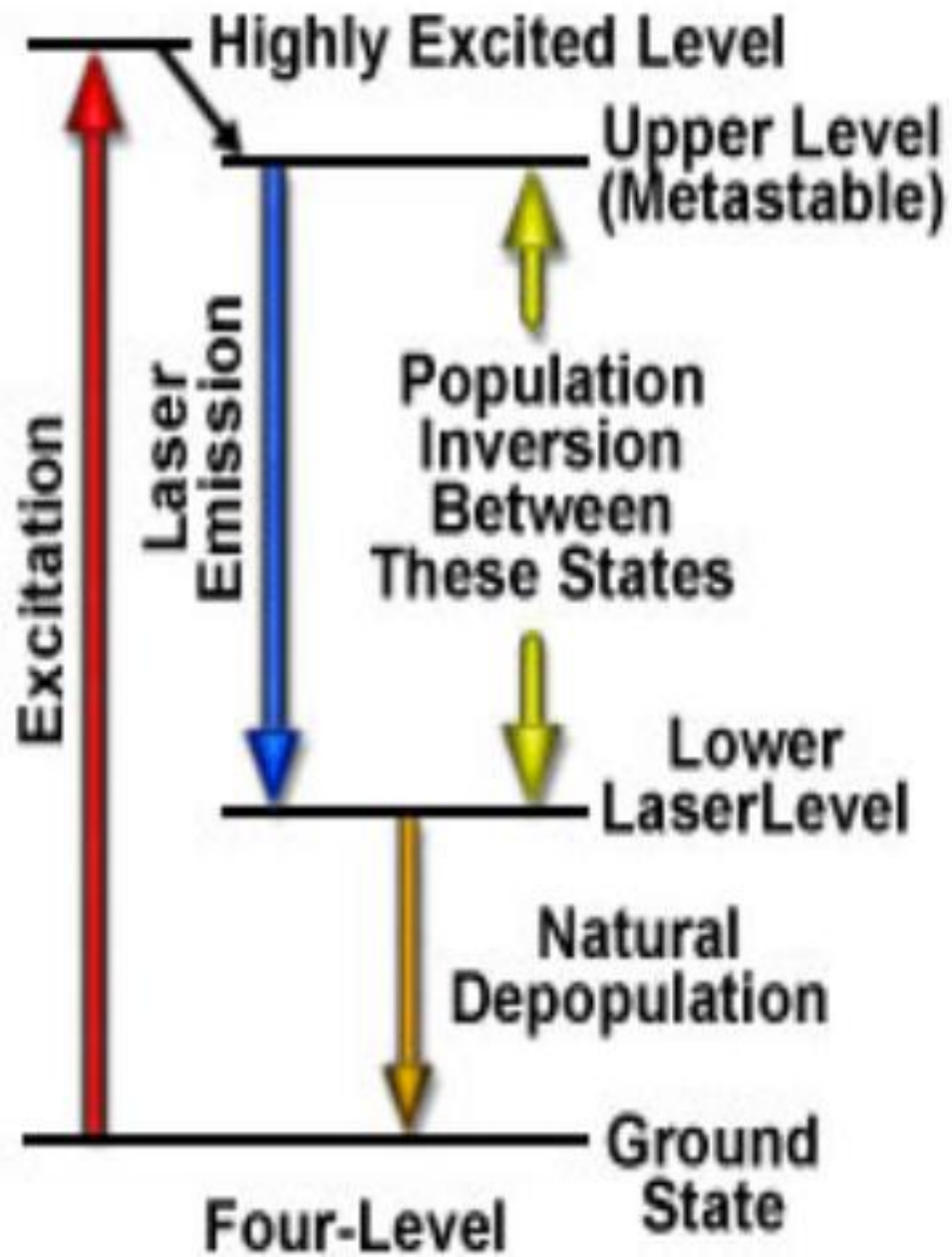


Figure 1



LASER OPERATION:

- **Active Medium:** Laser material such as Ruby Crystal, Gas mixture, Atoms, Molecules or Ions that emit light in visible and invisible regions.
- **Source of Excitation:** Optical (Flash Lamp), Electric discharge, Thermal, Chemical etc.
- **Metastable State:** Higher energy state of lifetime (10^{-3} sec) between upper energy level (E_2) and lower energy level (E_1), in which atoms remain for a greater lifetime than (E_2). The transition between Metastable state and lower energy state (E_1) produces stimulated emission.
- **Population inversion:** The interchange of the number of atoms in upper energy state of lifetime (10^{-7} sec) and the number of atoms in metastable state (10^{-3} sec), i.e. more atoms in excited state than in ground state.
- **Feedback:** Motion of Particles back and forth.

energy of
atom

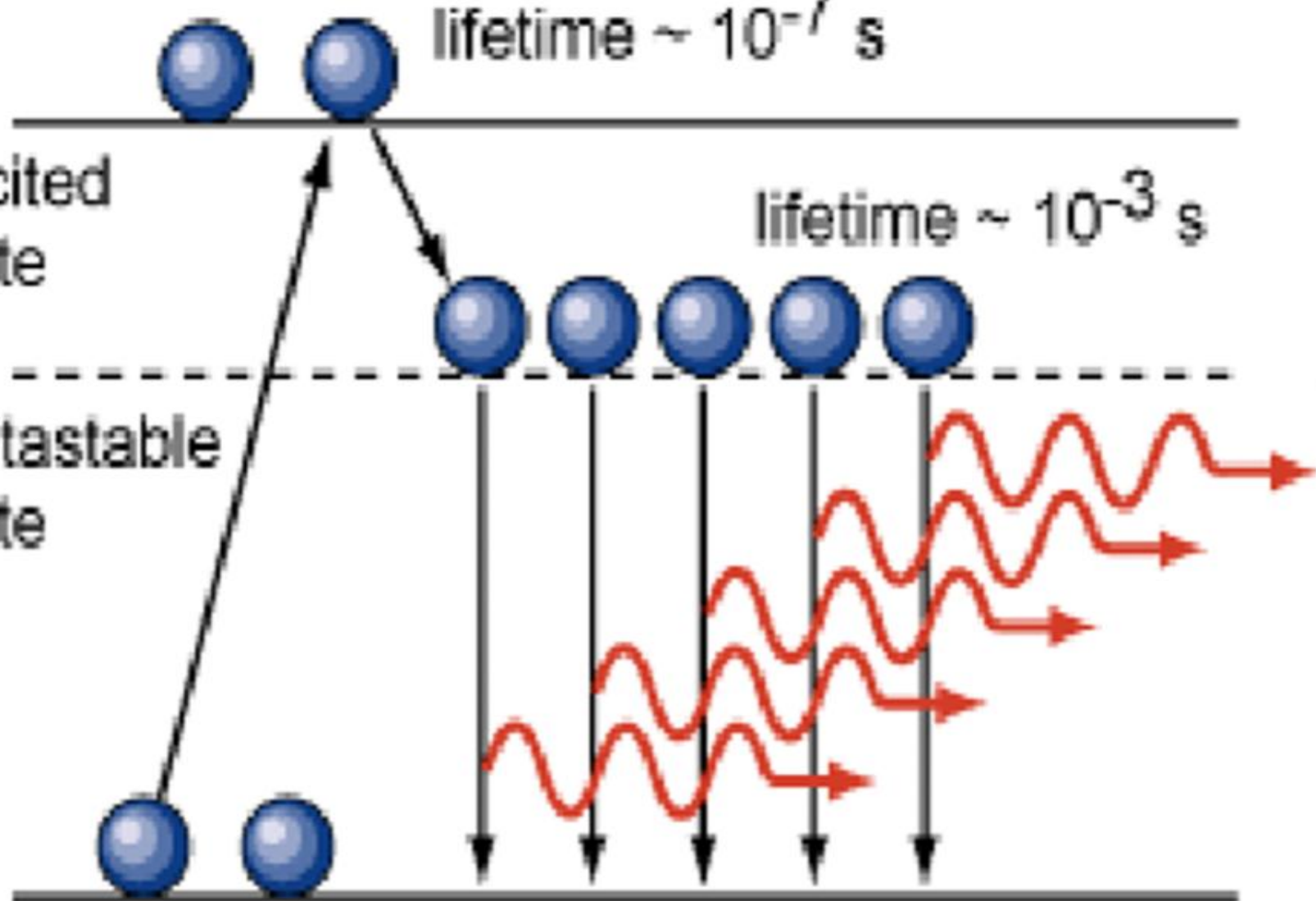
lifetime $\sim 10^{-7}$ s

excited
state

lifetime $\sim 10^{-3}$ s

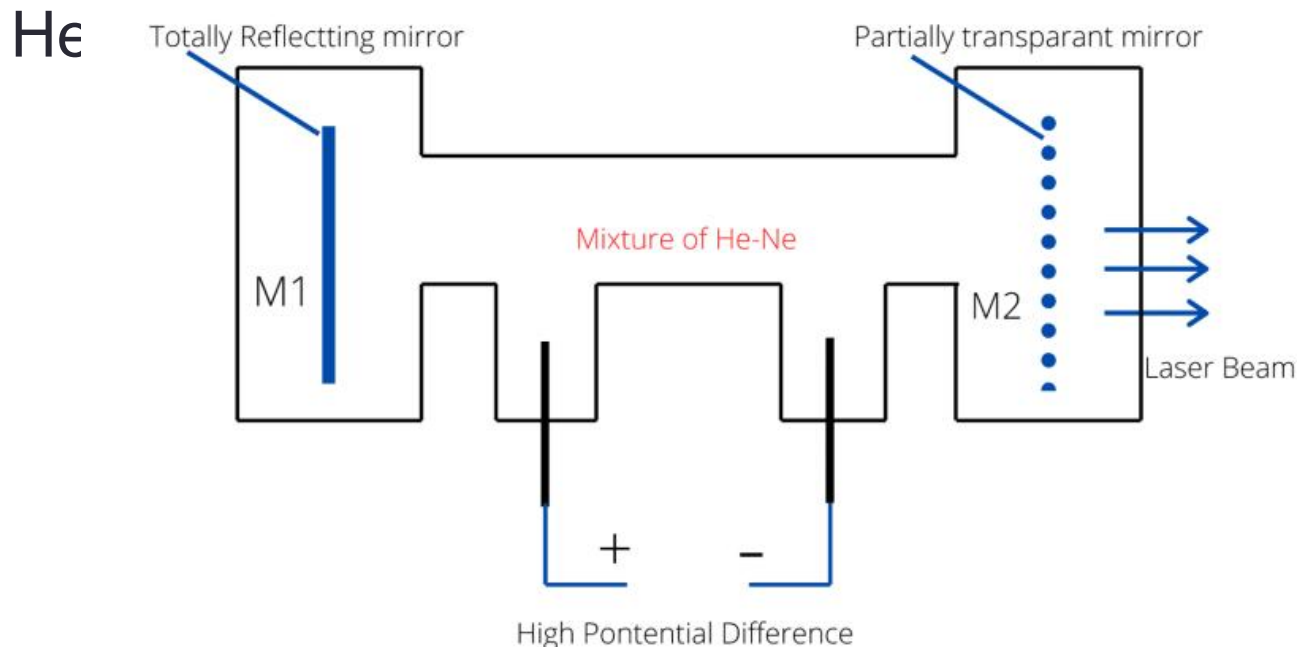
metastable
state

ground
state



He-Ne LASER:

- Helium-Neon laser is a type of **gas laser** in which a mixture of helium and neon gas is used as an active or gain medium and electrical pumping is used to achieve population inversion. Helium-Neon laser is also known as



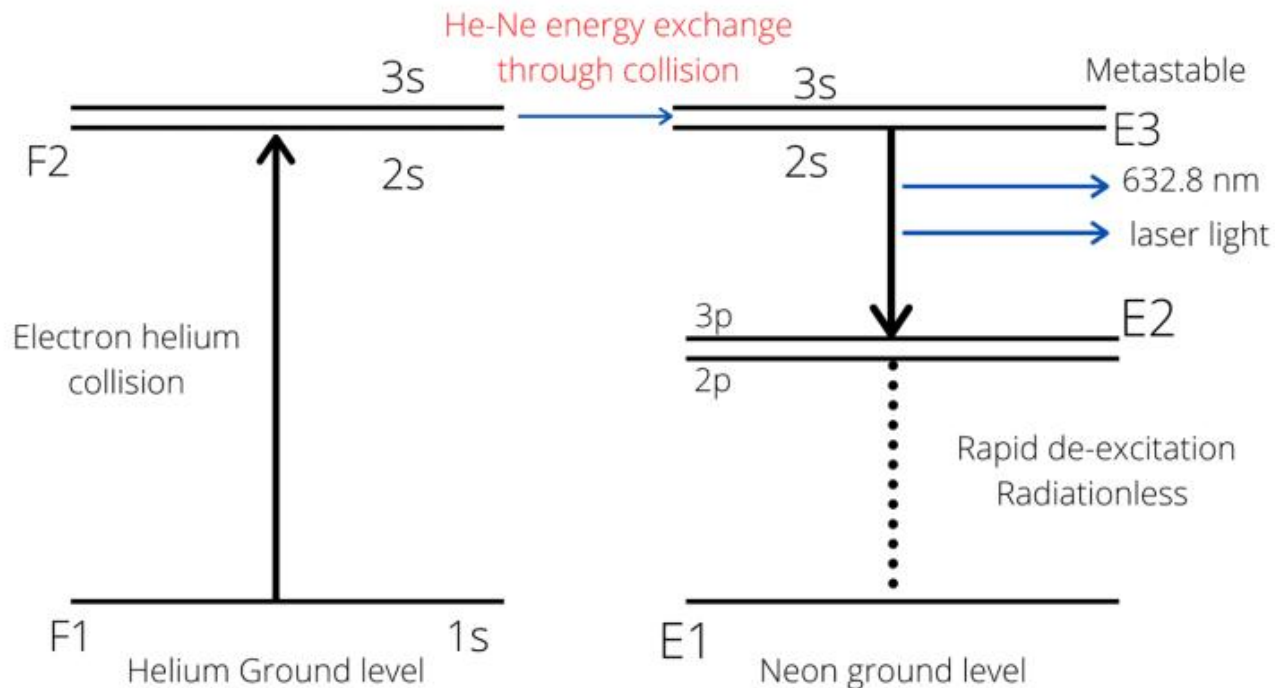
Construction of He Ne Laser;

- The Helium-neon gas laser system consists of a glass tube of nearly one meter in length. The gain medium in the He-Ne laser is a mixture of helium (90%) at 1 mm of Hg pressure and neon (10%) at 1/10 mm of Hg pressure filled in the glass tube.
- The reflecting mirrors M1 and M2 are fixed at the end of the tube. The mirror M2 is partially transparent and the laser light comes out of it. On the other hand, the mirror M1 is totally reflecting mirror.
- The glass tube containing the He-Ne gases mixture and has two mirrors at its end is called a **resonator** or **optical cavity**.
- A high potential difference is applied across the two electrodes of the discharge tube for creating population inversion.

Helium-Neon laser Working:

- Helium and neon are noble gases having only one atom per molecule. The energy level diagram of helium and neon atom is shown below.

He-Ne laser energy level diagram



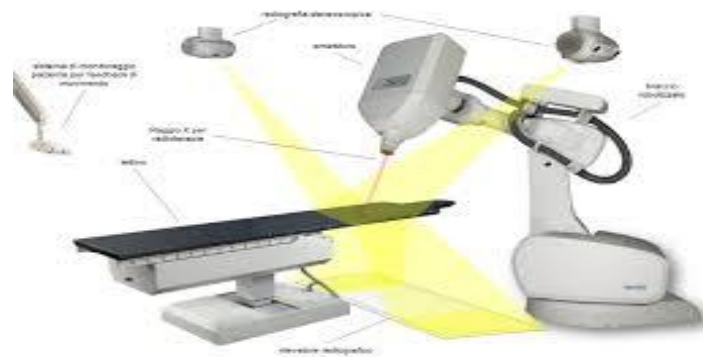
- When electric discharge occurs in the discharge tube, some of the atoms of the mixture of **He-Ne** that ionized. The electrons obtained so collide with Helium atom in the ground state or level and hence Helium atom goes to the excited state F2.
- The helium atom requires less [energy](#) to go to the excited state F2 from its ground level or state than and the Neon atom to go to the excited state E3 from its ground level or state ($E3 > F2$). The Helium atom at the excited state (F2) is Forbidden to make a spontaneous transition to the ground state (F1).
- The excited Helium atom (at level F2) collides with unexcited neon atoms at the ground state. The excitation energy (E3) of neon atoms nearly matches the excitation energy (F2) of the Helium atom.
- Therefore, there is an exchange of energy among the atom of helium in the excited state F2 and the atoms of neon gas in the ground state.
- As a result of this, the Helium atoms fall back to the ground state and the neon gas neon atom goes to the excited state E3 whose lifetime is greater than the lifetime of the energy state E2 and hence called **metastable state**. Thus, the Helium atom in the excited state act as a pumping source for a neon atom.

- Neon atom in the metastable state any E3 jumps to the lower energy state E2 through spontaneous emission and a Photon of wavelength 632.8 nm is emitted due to the transition 3s – 2p, This is the most prominent wavelength in He Ne laser in the red portion of the spectrum.
- The other wavelength emitted in Helium-neon laser is 3.39 μm due to the transition 3s – 3p and 1.15 μm due to the transition 2s – 2p.
- The Neon atom in the excited state E2 (having a small lifetime) rapidly deexcited to the Neon ground state by the non-radioactive transition. Thus, there are more neon atoms in the metastable state E3 than the atom in the excited state E2.
- Hence, population inversion is established between the energy states E3 and E2. The population inversion of a neon atom is achieved through Helium atoms.

Characteristics Helium Neon Laser:

Sr. No.	Title	Description
1.	Type	Gas Laser.
2.	Active Medium	A mixture of helium and neon gas.
3.	Pumping Method	Electrical pumping.
4.	Power output	He-Ne laser generates power of less than 10mW.
5.	Wavelength of output	The most common wavelength produced by the He-Ne laser is 632.8 nm.

APPLICATIONS OF LASER



Lasers in Science and Technology:

1. A laser helps in studying the Brownian motion of particles.
2. [Helium-neon](#) laser, proved that the velocity of light is same in all directions.
3. With the help of a laser, it is possible to count the number of [atoms](#) in a substance.
4. In computers to store and retrieve information from a Compact Disc (CD).
5. To measure the pollutant gases and other contaminants of the atmosphere.
6. Determine the rate of rotation of the earth accurately.
7. Used in computer printers.
8. Used for producing three-dimensional pictures in space without the use of lens.
9. Used for detecting earthquakes and underwater nuclear blasts.
10. A gallium arsenide diode laser can be used to setup an invisible fence to protect an area.

Lasers in Medicine:

1. Tumors, ulcers and cancer diagnosis and treatment.
2. For eye lens curvature corrections.
3. To study the internal structure of microorganisms and cells.
4. To produce chemical reactions.
5. To create plasma.
6. Remove the caries or decayed portion of the teeth.
7. In cosmetic treatments.

Lasers in Communications:

1. Optical fiber communications.
2. Underwater communication networks.
3. Space communication, radars and satellites.

Lasers in Military:

1. Laser range finder.
2. Ring laser gyroscope.
3. Laser Guided Anti-Tank-Missile (ATM).
4. Laser Radar (Lidar).
5. Air Reconnaissance.

Lasers in Industries:

1. To cut glass and quartz.
2. In electronic industries for trimming the components of Integrated Circuits (ICs).
3. For heat treatment in the automotive industry
4. Bar code scanning.
5. To drill aerosol nozzles and control orifices.